

FORM 2
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COMPLETE SPECIFICATION
(See section 10 and M.13)

AI-PRO-HELMET

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The following specification particularly describes the invention and the manner in which it is to be performed.

AI-PRO-HELMET

FIELD OF INVENTION

The field of invention is primarily relates to the field of safety engineering, materials science and within the domains of Artificial Intelligence (AI), Machine Learning (ML), Computer Vision (CV) and Internet of Things (IoT). The present invention relates to intelligent rider safety systems, more particularly to a smart helmet integrating real-time monitoring, transparent display technology, autonomous vehicle control, alcohol-based ignition prevention, navigation assistance, and emergency response features.

BACKGROUND OF INVENTION

Two-wheeler riders face significantly higher risk of accidents and fatalities due to lack of protective and assistive technologies. Existing helmets provide basic head protection but lack intelligent features such as crash detection, fatigue monitoring, speed regulation, and emergency communication. In addition, riders frequently experience heat buildup inside helmets, especially in warm climates or during long rides. Traditional helmets do not include any active cooling mechanism, leading to discomfort, reduced concentration, and increased fatigue, which can further compromise rider safety. Similarly, existing vehicle safety systems do not enforce mandatory helmet usage, nor do they incorporate alcohol detection directly into the rider's wearable equipment. Navigation devices commonly installed on motorcycles may distract the rider by obstructing vision or requiring manual interaction. There is therefore a need for a comprehensive safety ecosystem that integrates helmet-based sensing, autonomous vehicle control, transparent displays, communication systems, environmental monitoring, liquid-based cooling, and ignition control to ensure maximum rider protection, comfort, and enhanced situational awareness.

SUMMARY OF INVENTION

The present invention provides an integrated smart helmet system (100) designed to enhance rider safety, comfort, and vehicle control through a series of intelligent modules. The system comprises: (101) a transparent rollable display for real-time navigation, rider information, real-time traffic and real-time advertisement ; (102) an alcohol detection sensor linked to the vehicle's ignition system to prevent drunk driving; (103) a crash detection module capable of identifying impacts and automatically triggering emergency communication; (104) a fatigue and sleep-detection unit that monitors eye activity and issues alerts or initiates vehicle slowdown if the rider becomes unresponsive; (105) an automatic speed-control mechanism that regulates vehicle speed based on road limits and map data; (106) an integrated indicator and brake-light module to enhance rider visibility; (107) Integrated brake lights and turn indicators consist of high-visibility LED arrays embedded in the helmet shell and connected to the helmet controller so they can be synchronized with vehicle signaling or activated directly by the rider through gestures, voice, or automated events (e.g., emergency braking).; (108) a solar charging unit for sustainable power supply; (109) detachable hardware components that allow easy maintenance and portability; (110) a mandatory helmet-vehicle integration system

that prevents vehicle ignition unless the helmet is detected and authenticated; (111) voice command and eye-gesture control for fully hands-free operation; and (112) a liquid cooling system employing microfluidic channels and temperature-controlled coolant circulation to maintain comfortable internal temperature during extended or high-heat riding conditions. Together, these components create a comprehensive intelligent safety ecosystem that ensures rider protection, improves comfort, and integrates seamlessly with vehicle operation. (113) Data Collection Model is trained to identify speed-limit signs under different lighting, weather, Data is used to provide better service to overall community of users.

DETAIL DESCRIPTION

The invention is an integrated smart helmet system (100) combining multiple interdependent modules to maximize rider safety, comfort, and vehicle control: (101) a transparent rollable display is embedded within the visor and constructed from flexible, see-through display technology so that navigation, speed, incoming alerts, and system status are presented directly in the rider's line of sight without obscuring vision; the display mechanically or electro-active rolls away when not required to preserve an unobstructed field of view. (102) An alcohol detection subsystem, positioned in the chin or mouth region, continuously samples exhaled breath using an electrochemical or fuel-cell sensor and communicates its measured alcohol concentration to the helmet's controller and the vehicle ignition unit via a secure wireless link; when the measured level exceeds a configurable safety threshold the system issues warnings and prevents engine start or engagement until cleared. (103) The crash detection system integrates tri-axial accelerometers, gyroscopes, and high-g impact sensors with on-board signal processing to identify collision signatures (sudden deceleration, rotational spikes, abnormal orientation changes) and, upon detection, automatically initiate emergency procedures including locking GPS coordinates, broadcasting an emergency message to preconfigured contacts and authorities, and enabling remote telemetry for first responders. (104) Fatigue and sleep detection is implemented using infrared eye-tracking and head-pose sensors that monitor blink rates, eyelid closure duration, micro-saccades, and nodding behaviour; the module generates graded alerts (visual, auditory, haptic) when drowsiness is detected and, if no rider response is observed, escalates by commanding the vehicle to slow and eventually bring the vehicle to a safe stop through the mandatory vehicle-helmet control channel. (105) The smart speed controller fuses GPS-derived location, map metadata, and regulatory speed limit feeds with real-time vehicle speed telemetry to apply corrective commands to the vehicle's drive system or to provide progressive alerts, thereby enforcing compliant speeds automatically or assisting the rider in maintaining safe velocity for the prevailing road segment. (106) Navigation assistance comprises an integrated GPS module and map stack that renders route guidance, turn-by-turn instructions, and traffic alerts on the transparent display; the navigation UI is controllable via voice commands or eye gestures to minimize manual interaction and cognitive load. (107) Integrated brake lights and turn indicators consist of high-visibility LED arrays embedded in the helmet shell and connected to the helmet controller so they can be synchronized with vehicle signaling or activated directly by the rider through gestures, voice, or automated events (e.g., emergency braking). (108) A solar charging subsystem uses thin-film photovoltaic cells mounted on exposed helmet surfaces to trickle-charge the internal battery, extending operational autonomy and enabling sustainable continuous monitoring; energy management circuitry prioritizes critical safety subsystems and charges detachable

battery modules. (109) All active subsystems—display, sensors, cooling unit, battery, and communications modules—are designed as detachable, portable units with standardized connectors and secure mechanical mounts to facilitate maintenance, upgrades, or replacement without specialized tools. (110) Mandatory integration for ignition is accomplished through an authenticated, encrypted handshake between the helmet and the vehicle engine control unit; the vehicle will not enable ignition or transmission unless the helmet is present, verified, and in an operational state, thereby enforcing helmet use and system readiness. (111) Voice and eye-gesture control provide hands-free human-machine interaction: a far-field microphone array supports natural language voice commands and noise-robust speech recognition, while eye-tracking combined with dwell/gesture logic allows selection and navigation of on-screen elements without manual input. (112) The liquid cooling system consists of microfluidic channels embedded in the inner lining, a miniature centrifugal or peristaltic pump to circulate coolant from a detachable reservoir, a thermoelectric cooler or compact heat-exchanger to dissipate heat from the circulating fluid, and temperature sensors that manage on/off control and pump speed; the cooling loop draws power from the helmet battery or solar supply and is designed for low weight, minimal noise, and modular removal for cleaning or refill—together these features reduce internal heat buildup, improve comfort and concentration, and thereby reduce heat-related fatigue that can impair rider safety. (113) Data Collection Model it consist of A camera-based data acquisition and traffic-intelligence module incorporates a compact front-mounted camera that continuously captures real-world riding footage; onboard algorithms process the video stream to detect traffic density, identify road hazards, classify speed-limit signboards, and track lane conditions. The extracted metadata, along with periodic image samples, is transmitted securely to a cloud or edge server for data logging, large-scale traffic analysis, route optimization, and system performance improvements. When speed-limit signs are recognized, the helmet compares the detected limit with live vehicle speed and communicates alerts or regulatory commands to the vehicle controller to ensure speed compliance, thereby merging real-time perception, cloud analytics, and intelligent vehicle control into a unified safety ecosystem, Data is used to provide better service to overall community of users .

DETAIL DESCRIPTION OF DRAWINGS

An exemplary embodiment of figure 1 is illustrated.

An exemplary embodiment of figure 2 is illustrated in a flowchart depicting workflow.

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WE CLAIM:

1. A AI-PRO-HELMET (100) comprising:
 - i. A roll-able integrated dashboard display module (101) configured to present navigation data, speed information, system alerts, and vehicle-status indicators using a transparent, flexible, and visor-embedded display substrate, the module further configured to mechanically or electro-actively retract into a rolled state when not required to maintain an unobstructed field of view;
 - ii. A liquid-cooling and thermal-comfort module (112) including microfluidic heat-exchange channels, a miniature circulation pump, a coolant reservoir, and temperature sensors, all configured to circulate coolant through the helmet lining and regulate internal temperature to reduce heat accumulation and rider fatigue;
 - iii. A speed-compliance and limiter-analysis module (113) comprising a real-time speed interpreter, sign-recognition logic, and vehicle-telemetry integration, the module configured to identify regulatory speed-limit signboards, compare detected limits with live vehicle speed, and issue alerts or corrective control commands to enforce or assist in speed compliance;
 - iv. An integrated navigation-map assistant module (106) incorporating a GPS receiver, map-rendering engine, and a voice- or eye-gesture-controlled user interface, configured to provide turn-by-turn guidance, route planning, traffic updates, and adaptive navigation instructions on the visor-embedded display with minimal manual interaction;
 - v. A communication module (109) configured to establish secure data transfer between the helmet, vehicle, and remote servers, the module supporting the transmission of sensor data, traffic metadata, system diagnostics, and user-authorized control commands to enable coordinated vehicle-helmet operation and cloud-based analytics.
2. The AI-PRO-HELMET as claimed in claim 1, wherein the transparent rollable display module (101) comprises flexible OLED or polymer-based technology capable of retracting into the visor frame when not in use.
3. The AI-PRO-HELMET as claimed in claim 1, wherein the alcohol detection module (102) comprises an MQ-3 or equivalent ethanol sensor calibrated for direct breath sampling.
4. The AI-PRO-HELMET as claimed in claim 1, wherein the crash detection module (103) automatically sends real-time coordinates, impact data, and rider medical ID profiles to emergency contacts or authorities.
5. The AI-PRO-HELMET as claimed in claim 1, wherein the fatigue detection module (104) issues visual, audio, and vibration alerts prior to activating vehicle slowdown protocols.
6. The AI-PRO-HELMET as claimed in claim 1, wherein the traffic sign recognition module (113) employs a convolutional neural network (CNN) trained to identify speed-limit signs under different lighting, weather, and motion conditions.

7. The AI-PRO-HELMET as claimed in claim 1, wherein the smart speed control module (105) restricts vehicle acceleration or automatically reduces speed when detected speed limits are lower than current vehicle speed. This module comprising SENSITIVITY ADJUSTABLE SENSOR BREAK AND CLUTH SYSTEM (SAS BREAKS AND CLUTH);
8. The AI-PRO-HELMET as claimed in claim 1, wherein the navigation assistance module (106) supports multimodal inputs including voice commands, eye gestures, and head movements.

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Total number of Sheets: 03

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ECOASTER x AI-PRO-HELMET (100)

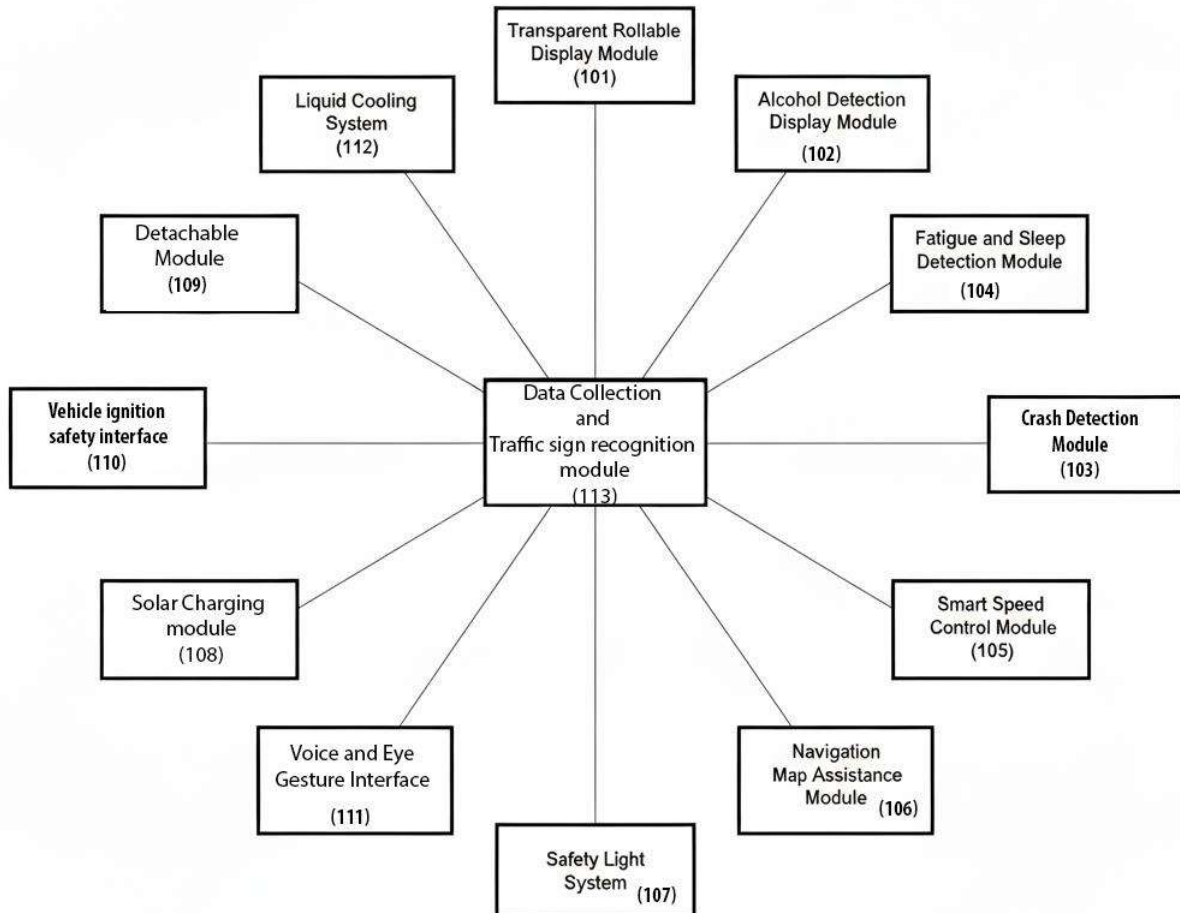


Fig1

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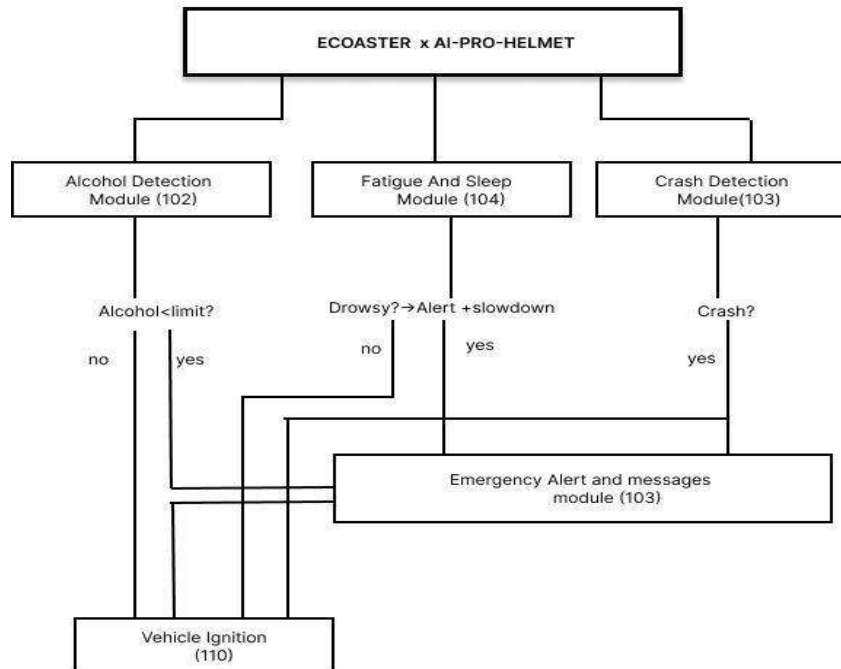


Fig 2

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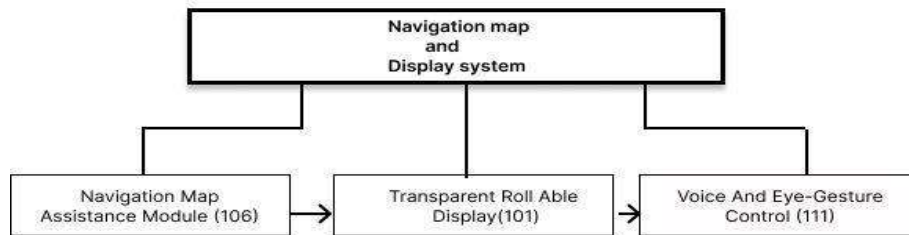


Fig 3

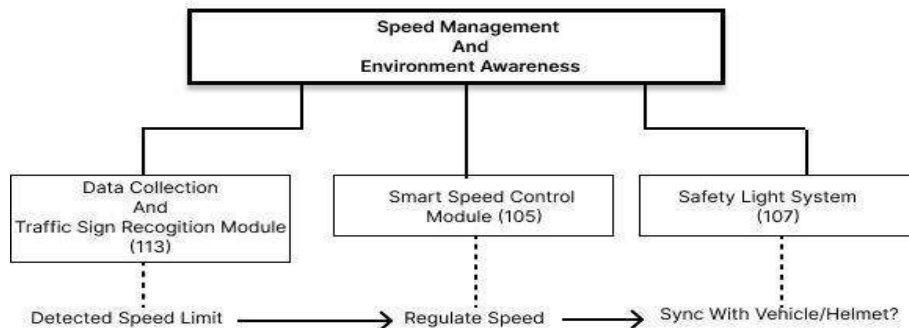


Fig 4

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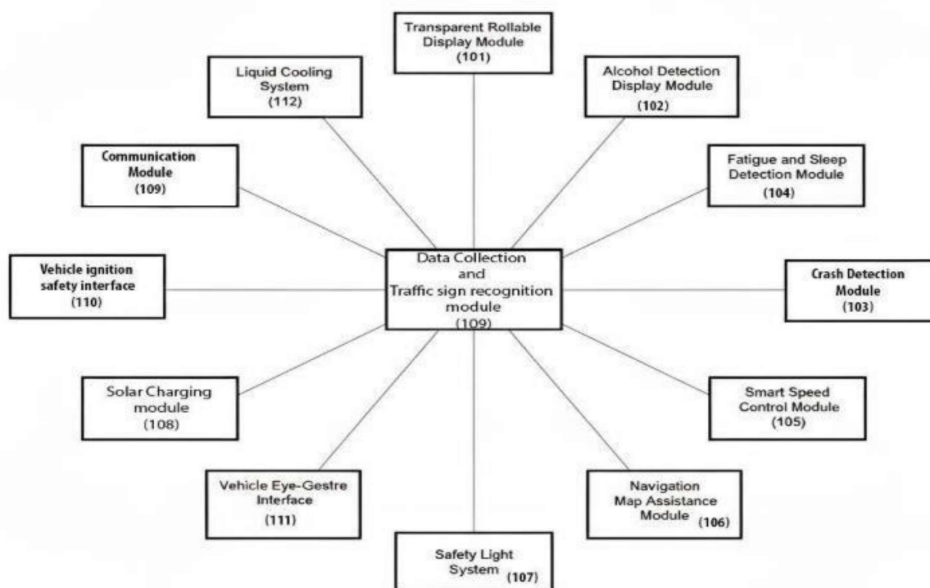
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ABSTRACT OF THE INVENTION

The invention provides an integrated smart helmet system that enhances rider safety, comfort, and vehicle compliance through advanced sensing and automated control. The helmet incorporates a transparent rollable display for unobstructed navigation, an alcohol sensor that restricts vehicle ignition, a crash detection unit that triggers automatic emergency alerts, and an infrared fatigue- monitoring system that initiates warnings or autonomous vehicle slowdown. A smart speed controller regulates vehicle speed based on GPS, map data, and camera-detected road signs. Additional features include solar charging, embedded LED indicators, modular detachable components, and a liquid-cooling system for rider comfort. A camera-based perception unit collects traffic and sign-board data, analyzes road conditions, and transmits metadata to a server, where aggregated data improves system accuracy and enhances services for the broader riding community. The system enforces mandatory helmet use and provides a comprehensive, intelligent safety ecosystem for two-wheeler riders.

ECOASTER x AI-PRO-HELMET (100)



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